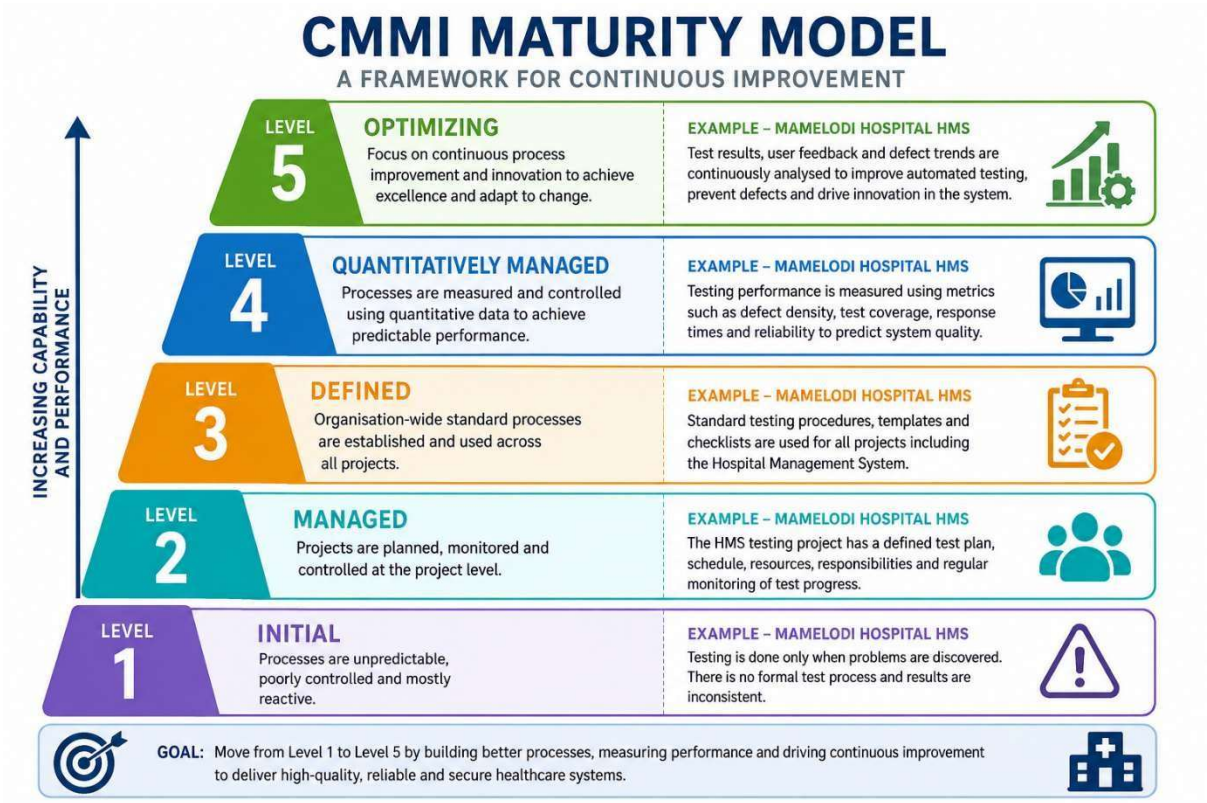
Question 1

Q1.1 Create an infographic illustrating the CMMI Model.

CMMI Maturity Levels Infographic Content

The Capability Maturity Model Integration (CMMI) provides a framework of best practices intended to improve organisational performance and critical processes (CMMI Institute, 2025).

According to the CMMI Institute, maturity levels provide an evolutionary path for organisational process and performance improvement. The five primary maturity levels range from Initial (Level 1) to Optimizing (Level 5).



Level 1: Initial is the lowest maturity level. At this level, processes are generally unpredictable, poorly controlled and reactive. Work is often performed without following formal procedures, and success may depend largely on individual e ort. In the Mamelodi Regional Hospital HMS project, this could mean that testing is only performed when problems are discovered, without a formal test plan or consistent testing process.

Level 2: Managed involves processes that are planned, monitored and controlled at the project level. Projects have clearly defined responsibilities, schedules and procedures. In the case of the Mamelodi Regional Hospital HMS, the testing project would have a formal test plan, a testing schedule, allocated resources and clearly assigned responsibilities for testers and developers.

Level 3: Defined means that processes are well documented, standardised and used consistently across the organisation. Instead of each project creating its own testing methods, the organisation follows

established procedures and standards. For Mint Management Group, this could involve using standard testing procedures, test case templates and checklists for all software projects, including the Hospital Management System.

Level 4: Quantitatively Managed focuses on measuring and controlling processes using quantitative data and performance metrics. The organisation uses measurable information to understand whether processes are performing e ectively and predict future performance. In the Mamelodi Regional Hospital HMS project, the testing team could measure defect density, test coverage, test execution results and system response times to evaluate the quality and performance of the system.

Level 5: Optimizing is the highest maturity level and focuses on continuous process improvement and innovation. Organisations continuously analyse their processes, identify weaknesses and introduce improvements to prevent future problems. For the Mamelodi Regional Hospital HMS, Mint Management Group could analyse test results, user feedback and defect trends to improve automated testing, prevent recurring defects and continuously enhance the quality of the system.

In conclusion, the five CMMI maturity levels provide a structured path for organisations to improve their processes. Mint Management Group should aim to progress towards the higher maturity levels to ensure that the development and testing of the Mamelodi Regional Hospital HMS are consistent, measurable and continuously improved. This will contribute to a reliable, secure and high-quality healthcare system (CMMI Institute, 2025).

3.2 Discuss the advantages of test automation in a software system, using the case study to illustrate your points.

Test automation involves using software tools to control test execution, compare actual results with expected results, establish test conditions and perform test reporting activities (Ammann and O utt, 2016).

Advantages of test automation for the Mamelodi Regional Hospital HMS

1. Faster execution of tests
   Automated tests can execute repetitive test cases faster than manual testing. For example, regression tests for patient registration can be run automatically whenever the system is updated.

2. Improved regression testing
   Changes to one module may a ect other modules. Automated regression tests can repeatedly check that registration, appointments, EMR, pharmacy and billing functions continue working correctly.

3. Consistency and reduced human error
   Automated tests execute the same steps consistently. This reduces variation caused by testers performing repetitive tasks di erently.

4. Supports continuous testing
   Automated tests can be executed regularly during development, allowing defects to be detected earlier.

5. Better use of human resources
   Testers can focus on complex activities requiring human judgement, such as exploratory testing and acceptance testing, while repetitive tests are automated.

6. Supports performance testing

Automation tools can simulate many users accessing the HMS simultaneously to determine whether the hospital system can handle periods of high demand.

Conclusion

For the hospital platform, test automation can improve efficiency and reliability. However, automation should complement rather than completely replace manual testing because usability and real healthcare workflows still require human judgement.

Q3.3 Discuss the importance of test documentation in the software testing process.

Test documentation is important because it:

1. Provides evidence of testing
   It shows what was tested, how it was tested and what the results were.
2. Improves traceability
   Requirements can be linked to test cases and defects. For example, the requirement that only authorised doctors may modify medical records can be linked to relevant security test cases.
3. Supports communication
   Test reports help project managers, developers and hospital stakeholders understand the quality and readiness of the HMS(IEEE, 2026; Spillner, Linz and Schaefer, 2014).
4. Supports future maintenance and audits
   Documentation can be used when the system is updated or investigated after deployment.

For Mamelodi Regional Hospital, proper test documentation is particularly important because the system manages sensitive patient information and supports critical hospital operations.

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